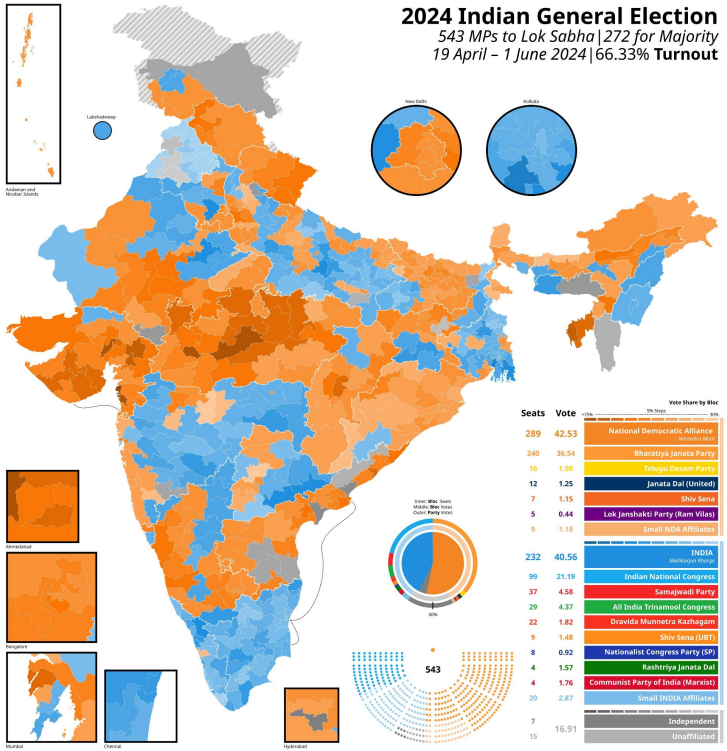
Introduction to Data Science

Manu Madhavan

Outline

Suppose I give you information about 5,000 Malayalam movies released over the last 50 years. What interesting questions can you answer?

Can we predict election results?



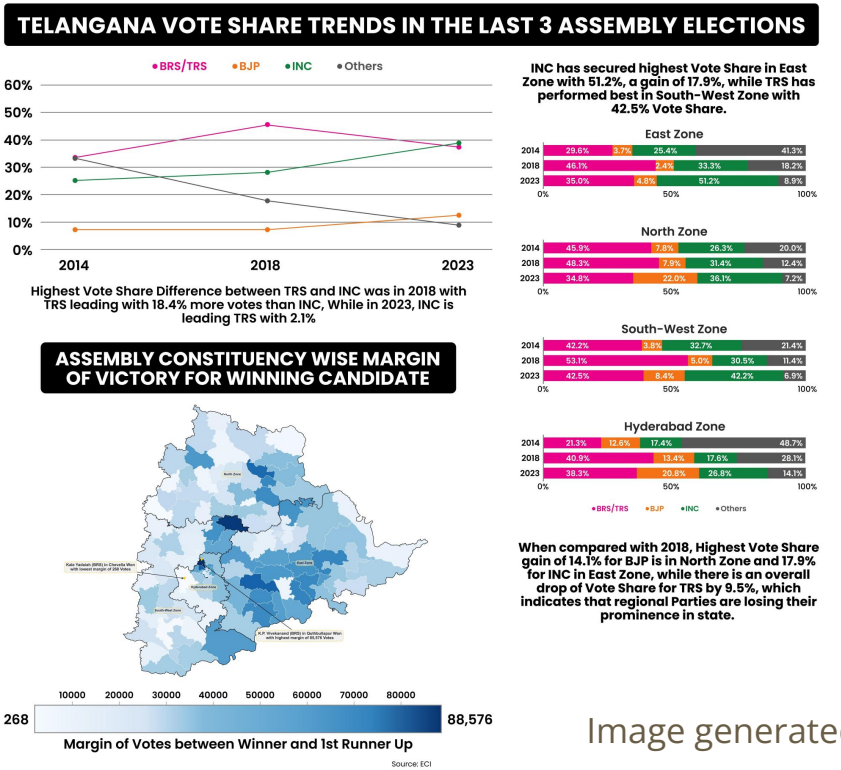
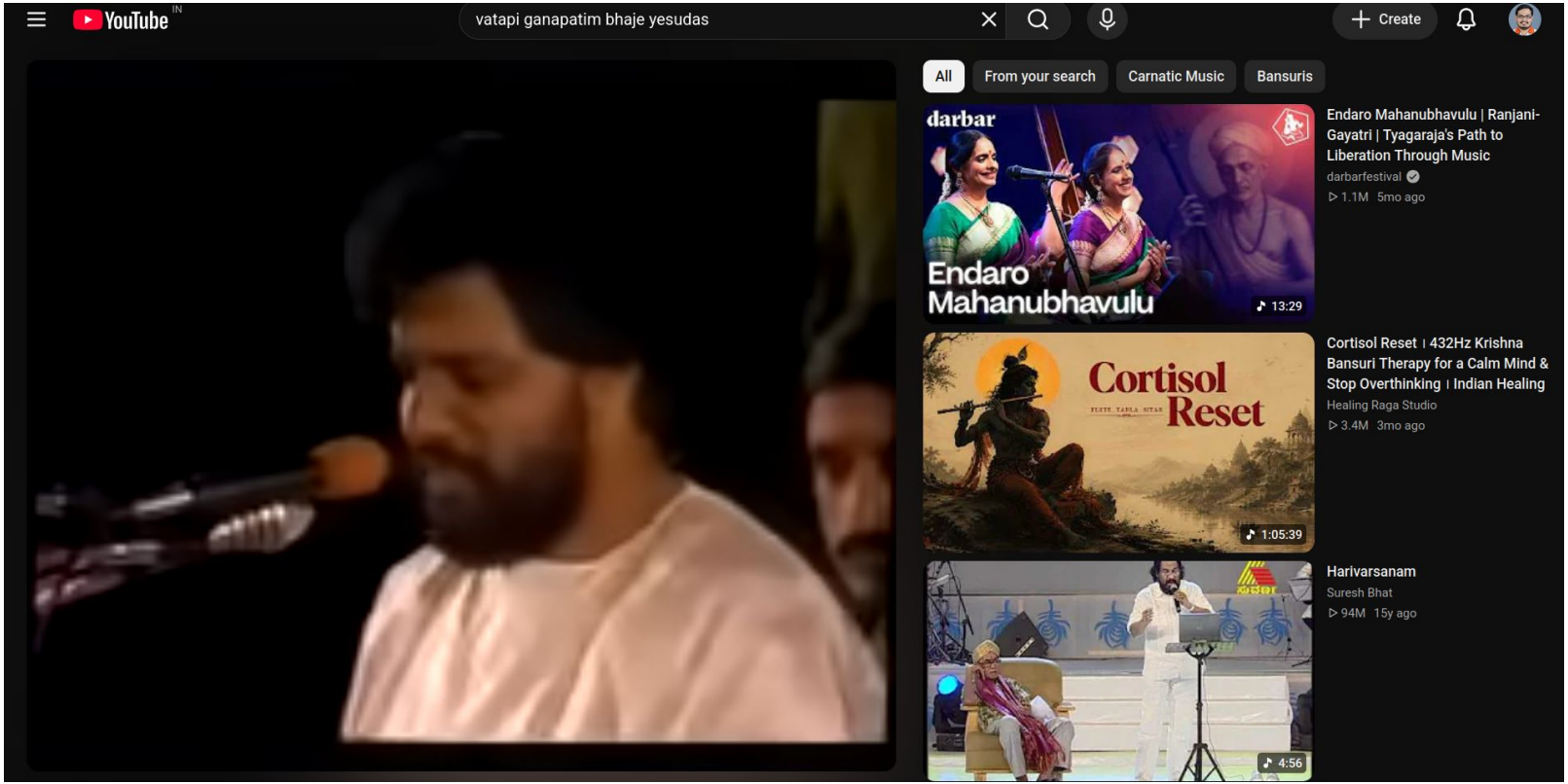


Image generated by Gemini

How does Netflix or Spotify or Youtube know what we like?



How IMD makes extreme rainfall prediction?

The Hindu

Cyclonic storm Arnab impact on Telangana | Heavy-very heavy rainfall forecast in four districts, yellow alert in four more on Oct. 28

Cyclonic storm Arnab may bring heavy to very heavy rainfall to four districts in Telangana and rainfall in four more on October 28, 2026,...

4 days ago



World Meteorological Organization WMO

El Niño set to become very strong, raising risks of extreme weather into 2027

Geneva, Switzerland (WMO) - El Niño is firmly established and will intensify into a very strong event in the coming months, with big impacts...

3 Sept 2026



www.pib.gov.in

IMD Forecasts Enhanced Rainfall Across East, Northeast India and Western Himalayan Region Over Next 7 Days; Extremely Heavy Rain Alert Issued for Odisha Today

IMD Forecasts Enhanced Rainfall Across East, Northeast India and Western Himalayan Region Over Next 7 Days; Extremely Heavy Rain Alert...

16 Jul 2026



Can Data Science help farmers decide when to sow?

Ministry of Agriculture & Farmers Welfare

Government Conducts AI-Based Pilot for Local Monsoon Forecasting to Support Kharif Sowing Decisions

Posted On: 17 MAR 2026 7:12PM by PIB Delhi

In the Department of Agriculture and Farmers Welfare, an AI-based pilot was conducted in collaboration with the Development Innovation Lab-India on agriculturally relevant local monsoon onset forecasts across parts of 13 states in India for Kharif 2025. An open-source blended model was used, including NeuralGCM, the European Centre for Medium-Range Weather Forecasts' (ECMWF) Artificial Intelligence Forecasting System (AIFS), and historical rainfall data from 125 years from the India Meteorological Department (IMD).

The probabilistic forecasts predicted only the local onset of the monsoon, which is essential for deciding on the date of sowing crops. Local monsoon onset forecasts were sent via SMS through the M-Kisan portal to 3,88,45,214 farmers in 13states in five regional languages- Hindi, Odia, Marathi, Bangla, and Punjabi.

Telephonic farmer feedback surveys were conducted in Madhya Pradesh and Bihar through Kisan Call Centers after the forecasts were sent. The survey revealed that 31% to 52% farmers adjusted their planting decisions, primarily through changes in land preparation and sowing timing, which included crop and input choice.

Top picks for you



Aviation Metal & Alloys Pure Titanium Wire 0.50mm x 5M For Medical Uses or High Strength...
4.2 13
₹701.00



Sabine's Notebook: In Which the Extraordinary Correspondence of Griffin & Sabine Continues (Griffin and Sabine)
4.7 167



Invento 1pcs Al Aluminium Alloy 2mm Plate/Sheet...
3.5 37
₹290.00
Prime FREE Delivery



IBELL Angle Grinder AG10-70, 850W, Copper Armature, Disc...
4.1 1,744
₹1,706.00
prime FREE Delivery



iBELL 200-89 Inverter ARC Compact Welding Machine...
4.5 1,723
₹5,393.00
prime FREE Delivery



GVD PVC & FR Insulated 2 Core 1mm Lenth-10Mtr, Flexible Copper Wires & Cables for...
3.5 12
₹572.00



TheGiftKart Transparent Crystal Clear Back Cover for Samsung...
4.2 6,571
₹199.00
prime FREE One-Day



I Am a Strange Loop
4.5 389



HUPSHY Samsung Galaxy M21 2021 Armour Back Cover Case |...
4.2 1,738
₹185.00
prime FREE Delivery



The Idea Factory: Bell Labs and the Great Age of American Innovation
4.5 565



Stookin N20 3.7V - 6V 100 RPM Micro Gear Reduction DC Motor with 30:1 Metal Gearbox For RC...
3.5 76
₹349.00



Metamagical Themas: Questing For The Essence Of Mind And Pattern
4.5 69

Many more…can you suggest?

What is common among these applications?

What is common among these applications?

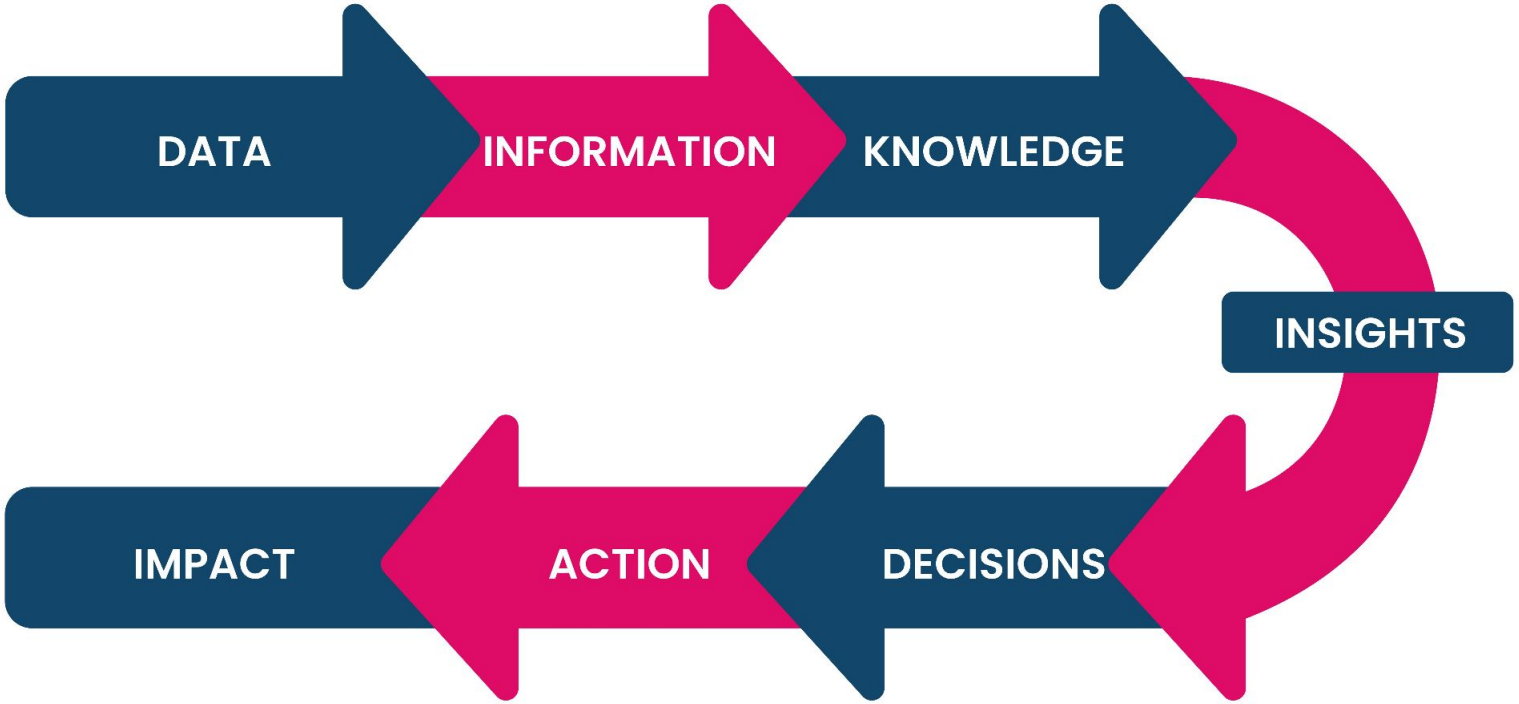


Image generated by ChatGPT

So, what is Data Science?

So, what is Data Science?

- An interdisciplinary field that uses mathematics, statistics, computer science, and domain expertise to extract meaningful insights and predictive patterns from raw data

- Data Ingestion → Data storage & Preprocessing → Data Analytics → Communicate

Why data science is important?

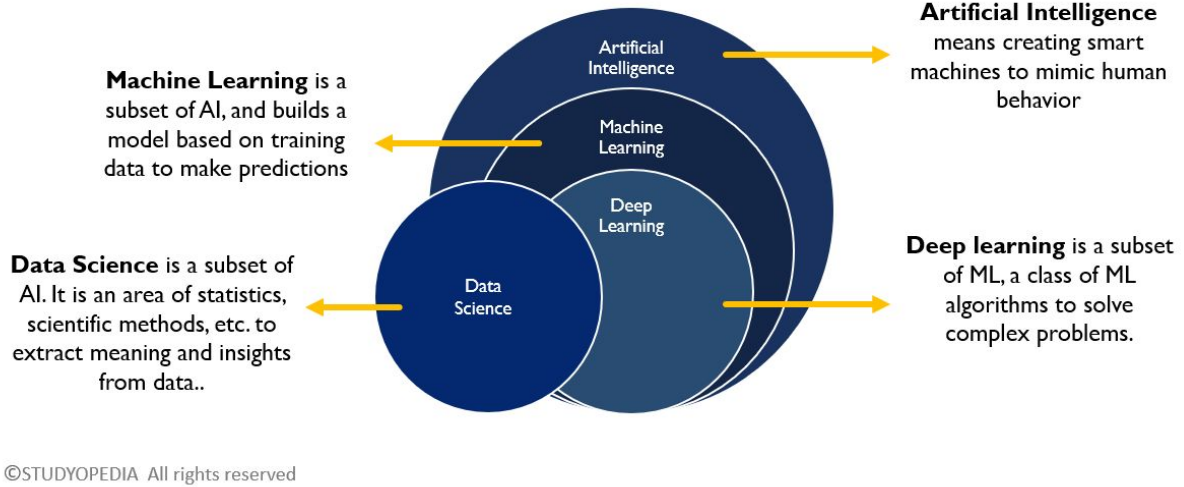
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Why data science?

- Data is everywhere!
- Predictive modeling to forecast future trends, risks, and customer behaviors
- Helps organizations to make Better Decisions
- Product Personalization

Burst the bubbles!

- AI
- Data Science
- Data Analysis
- Machine Learning
- Pattern matching
- Data Scientist



Fundamentals of Data Science

1. Understanding Data

- Types of data: structured, semi-structured and unstructured
- Data attributes: numerical, categorical, ordinal, etc.
- Data sources and quality

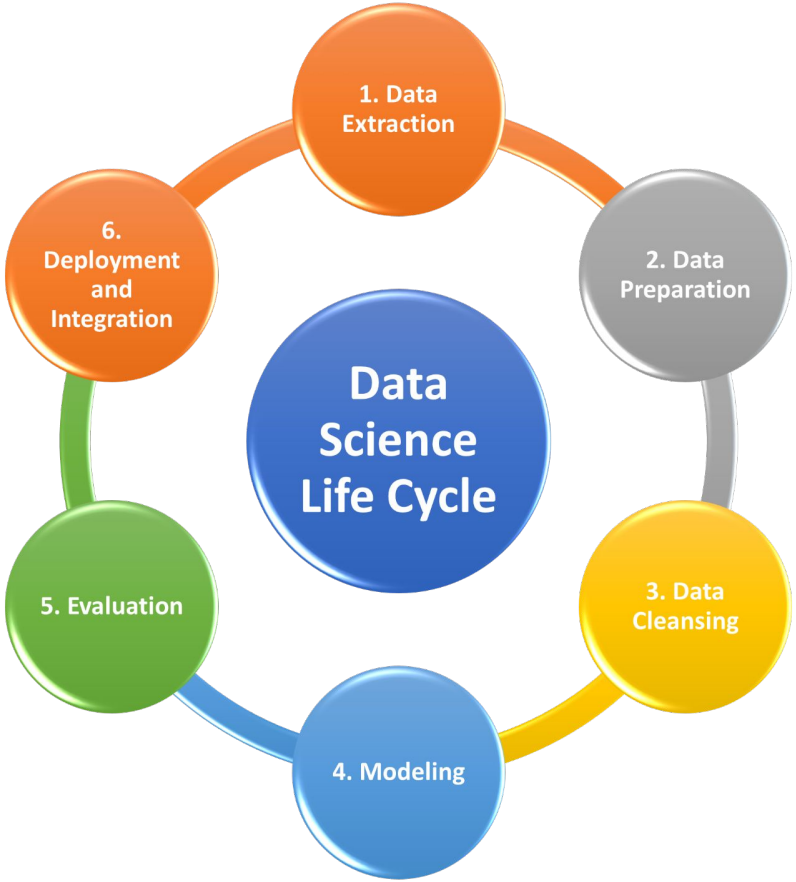
2. **Mathematical and Statistical Foundations**

- Basic statistics and probability
- Linear algebra and basic calculus (brief mention)
- Correlation and distributions

Foundation of Data Science

3. Data Science Life Cycle

- Problem definition
- Data collection and preparation
- Analysis and modelling
- Evaluation and communication



Foundation of Data Science

4. **Exploratory Data Analysis (EDA)**

   - Understanding patterns and distributions

   - Identifying missing values and outliers

   - Visualization and correlation analysis

5. **Descriptive Analysis**

   - What happened? What does the data tell us?

   - Summary statistics, trends and patterns

   - Dashboards and visual reports

Foundation of Data Science

6. Predictive Modeling

- What is likely to happen?
- Regression and classification
- Introduction to machine learning

So far...

- What is Data Science
- Why data science?
- How data science?
- Data Science life cycle
- Major components



Let's have a break now....

Exploratory Data Analysis

- EDA is the process of examining, summarizing and visualizing data to understand its characteristics, identify patterns, detect anomalies and generate hypotheses.

Before building a model, understand your data!

Why EDA?

- Understand the structure and characteristics of data.
- Identify missing values and inconsistencies.
- Detect outliers and unusual observations.
- Discover patterns and relationships.
- Identify useful features for modelling.

Types:
Univariate, Bivariate, Multivariate,

EDA Techniques

- Descriptive statistics
- Frequency distributions
- Histograms and box plots
- Bar charts and scatter plots
- Correlation analysis
- Pair plots

Python Libraries

- Pandas – data manipulation
- NumPy – numerical operations
- Matplotlib – visualization
- Seaborn – statistical visualization

Python Demo.

Summary

- Understand the data, variables
- Ask questions, visualize the data, observe and get insights

- Next: Use the questions and observations from EDA to decide what descriptive analysis or predictive modelling might be useful.

Lunch Break!

Descriptive Analysis

- Descriptive analysis involves summarising and presenting data to understand its main characteristics.
- Measures of Central Tendency: mean, median, mode
- Measures of Dispersion: Range, variance, SD, IQR
- Frequency Distribution and Grouping
- Data Visualization for Reporting

Python Demo

Predictive Modeling

- Predictive modelling uses historical data to develop a model that estimates unknown or future outcomes.
- Regression and Classification
- Data → Feature Selection → Train/Test Split → Model Training → Prediction → Evaluation
- Algorithms (linear regression, logistic regression, DT, RF,...)
- Model Evaluation
  - MAE, RMSE
  - Accuracy, Precision, Recall, F1 score, Confusion matrix

Hands-on: Titanic Survival Prediction

- Can we predict whether a Titanic passenger survived?
    - Input: Age, sex, passenger class, fare, etc.
    - Output: Survived (1) or did not survive (0).
    - Approach: Train a classification model and evaluate its performance.

Minor Assignment!

Explore a Malayalam Movie Dataset and Discover Interesting Patterns.

1. Identify three interesting questions.
2. Perform basic data cleaning.
3. Conduct EDA and create visualizations.
4. Present at least three observations.
5. Build a simple predictive model (optional).